Career Progression and Promotion Gap Analysis for Retention Optimization

A Data Analytics Project Using Power BI and Streamlit

Abstract

Employee retention is influenced by several workforce factors, including career progression, job role, workload, tenure and employee experience. This project develops a data-driven framework for examining promotion gaps and their association with employee attrition. A supplied HR analytics dataset containing 1,470 employee records and 31 variables was analyzed using Python for validation and Power BI for interactive reporting. The analysis focuses on YearsSinceLastPromotion, JobLevel, YearsAtCompany, YearsInCurrentRole, JobRole, OverTime, satisfaction measures and Attrition. The results show that the overall attrition rate in the supplied dataset is 16.12%. Employees working overtime have an observed attrition rate of 30.53%, compared with 10.44% among employees not working overtime. The findings are intended to support retention monitoring, career-path review and targeted workforce interventions. The analysis identifies associations only and does not establish causal relationships.

1. Introduction

Organizations need reliable workforce analytics to understand where employee turnover is concentrated and which career-progression indicators deserve attention. Promotion timing is particularly useful because long periods without advancement may indicate limited mobility, unclear career paths or a mismatch between employee expectations and available opportunities. This project therefore combines promotion-gap measures with attrition, job level, role and employee-experience variables to create an interactive management dashboard.

2. Objectives

1. Measure overall employee attrition and identify high-attrition workforce segments.
2. Analyze YearsSinceLastPromotion as a career-progression indicator.
3. Compare promotion gaps across job levels, departments and job roles.
4. Examine supporting retention factors such as overtime, satisfaction and tenure.
5. Build a Power BI dashboard for interactive analysis and a Streamlit dashboard for deployment.
6. Provide practical, evidence-based retention monitoring recommendations.

3. Dataset and Methodology

The supplied CSV contains 1,470 records and 31 columns. Key variables include Age, Attrition, Department, JobRole, JobLevel, MonthlyIncome, OverTime, PerformanceRating, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion and YearsWithCurrManager. Attrition is represented as a binary 0/1 field in the supplied file. Data preparation consists of type validation, aggregation and creation of measures for employee count, attrition rate, average tenure and promotion-gap indicators. A promotion-gap flag of 3 or more years is used as a dashboard segmentation rule; this is an analytical threshold for this project rather than a universal HR standard.

4. Power BI Dashboard Design

The Power BI report is organized into three pages. Executive Overview contains KPI cards and department/role comparisons. Promotion Gap Analysis focuses on YearsSinceLastPromotion, JobLevel and JobRole, with slicers for workforce segmentation. Retention Drivers compares overtime and employee-experience measures such as job satisfaction, environment satisfaction, work-life balance and job involvement. Recommended slicers include Department, JobRole, JobLevel, OverTime and Promotion Gap Band.

5. Results

The dataset contains 1,470 employees, of whom 237 are recorded as attrited, producing an overall attrition rate of 16.12%. Average age is 36.9 years, average monthly income is approximately 6,502.93, average tenure at the company is 7.0 years, and average years since the last promotion is 2.19. Sales Representative has an observed attrition rate of 39.76%, Laboratory Technician 23.94%, Human Resources 23.08%, and Sales Executive 17.48%. By department, Sales is 20.63%, Human Resources 19.05%, and Research & Development 13.84%. By job level, Level 1 is 26.34%, Level 2 is 9.74%, Level 3 is 14.68%, Level 4 is 4.72%, and Level 5 is 7.25%. Overtime is strongly associated with higher observed attrition: 30.53% for employees working overtime versus 10.44% for employees not working overtime.

6. Career Progression and Promotion Gap Interpretation

The dashboard should treat promotion delay as a monitoring signal rather than proof of a cause of attrition. In the supplied data, the average YearsSinceLastPromotion is 2.19 years. The relationship between promotion timing and attrition is not monotonic across every interval, so a single cutoff should not be interpreted as a causal threshold. Promotion analysis is most useful when combined with job level, role, tenure and employee-experience measures. Managers can use the report to identify groups where long promotion intervals coexist with elevated attrition and then review career pathways, internal mobility and development opportunities.

7. Retention Optimization Recommendations

1. Create transparent career-path criteria for each major job role.
2. Review employees with extended promotion intervals together with job level and tenure.
3. Track internal mobility, mentoring and training participation.
4. Monitor overtime as a retention signal, especially in groups with elevated attrition.
5. Use periodic Power BI reviews to compare attrition trends by role, level and department.
6. Validate dashboard findings with HR context before taking employee-level action.
7. Avoid using a single variable as a decision rule; combine multiple indicators and qualitative feedback.

8. Limitations

The supplied dataset is a cross-sectional HR analytics dataset and does not contain a time series of individual promotions or departures. It therefore cannot establish whether promotion gaps caused attrition. Some fields are categorical encodings, and the dataset's provenance should be verified before describing it as internal Palo Alto Networks employee data. The project uses a 3-year promotion-gap threshold for segmentation only. Results should be treated as analytical associations rather than causal or predictive conclusions.

9. Conclusion

This project demonstrates how Power BI can convert HR workforce data into an interactive career-progression and retention monitoring system. By combining attrition, promotion timing, job level, tenure, role and employee-experience indicators, organizations can identify workforce segments that warrant further review. The Streamlit version provides a lightweight deployment option, while Power BI provides interactive business reporting and filtering.

Keywords

Employee Attrition, Career Progression, Promotion Gap, Retention Analytics, Power BI, Streamlit, HR Analytics, Workforce Analytics

Appendix – Core Power BI Measures

```
Total Employees = COUNTROWS('Palo Alto Networks')
Attritions = SUM('Palo Alto Networks'[Attrition])
Attrition Rate = DIVIDE([Attritions], [Total Employees], 0)
Average Years Since Last Promotion = AVERAGE('Palo Alto Networks'[YearsSinceLastPromotion])
Promotion Gap Employees = CALCULATE([Total Employees], 'Palo Alto Networks'[YearsSinceLastPromotion] >= 3)
Promotion Gap Attrition Rate = DIVIDE([Promotion Gap Attritions], [Promotion Gap Employees], 0)
```